Supplementary Material

Competing-Risks Modelling of a Widely-Used Cirrhosis Outcome Benchmark: Recovering the Transplant Outcome and Cause-Specific Risk Lost to Classification

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These supplementary figures support the main text. Each is referenced there by its supplementary label (Fig. S1–S6). All are produced by the reproducible pipeline released with the paper, by `run_advanced.py` (S1), `figures_ext.py` (S2), `figures.py` (S3 to S5) and `run_finegray.py` (S6) in the code repository.

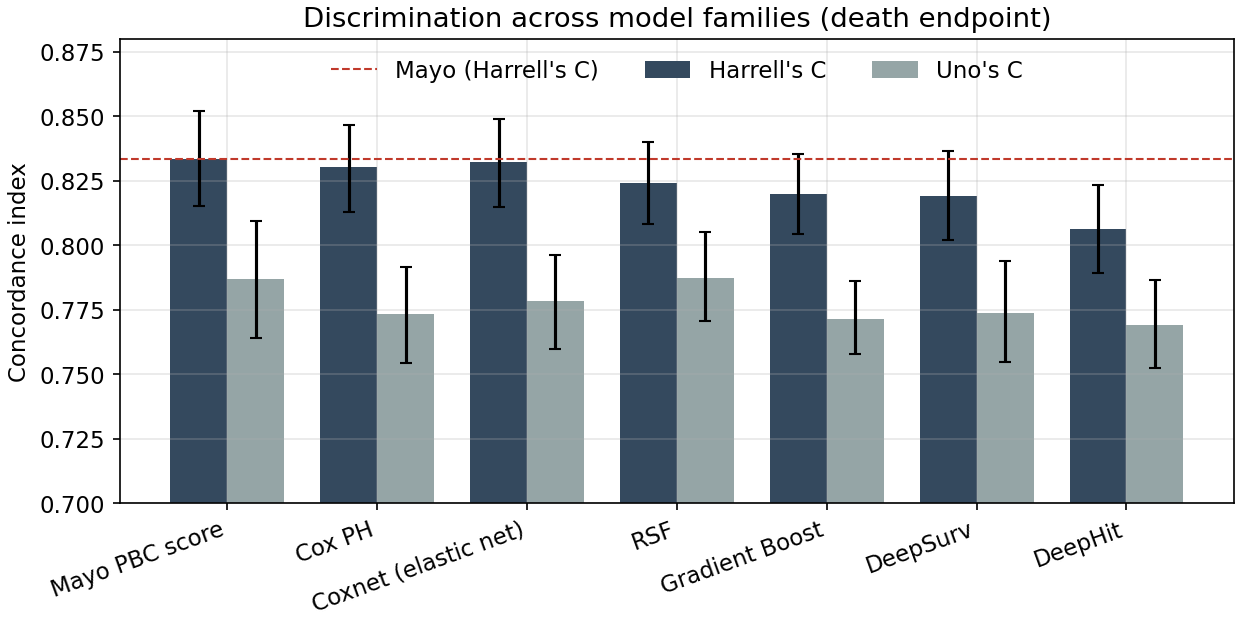


Figure S1: Concordance across model families for the death endpoint. The Mayo score (dashed line) is not exceeded by any ML model. (Companion to main-text Table 2, model comparison.)

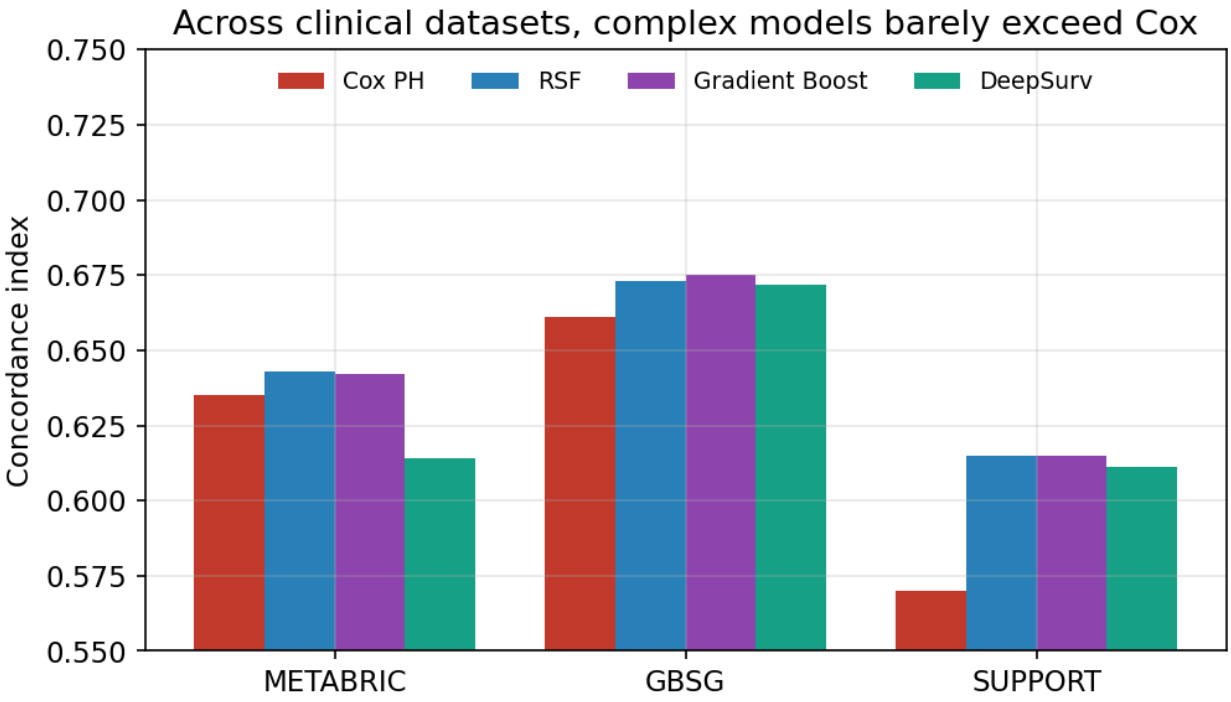


Figure S2: Across external datasets, complex models exceed Cox only marginally on smaller cohorts (METABRIC, GBSG) and meaningfully only on the largest (SUPPORT). (Companion to the external multi-dataset table.)

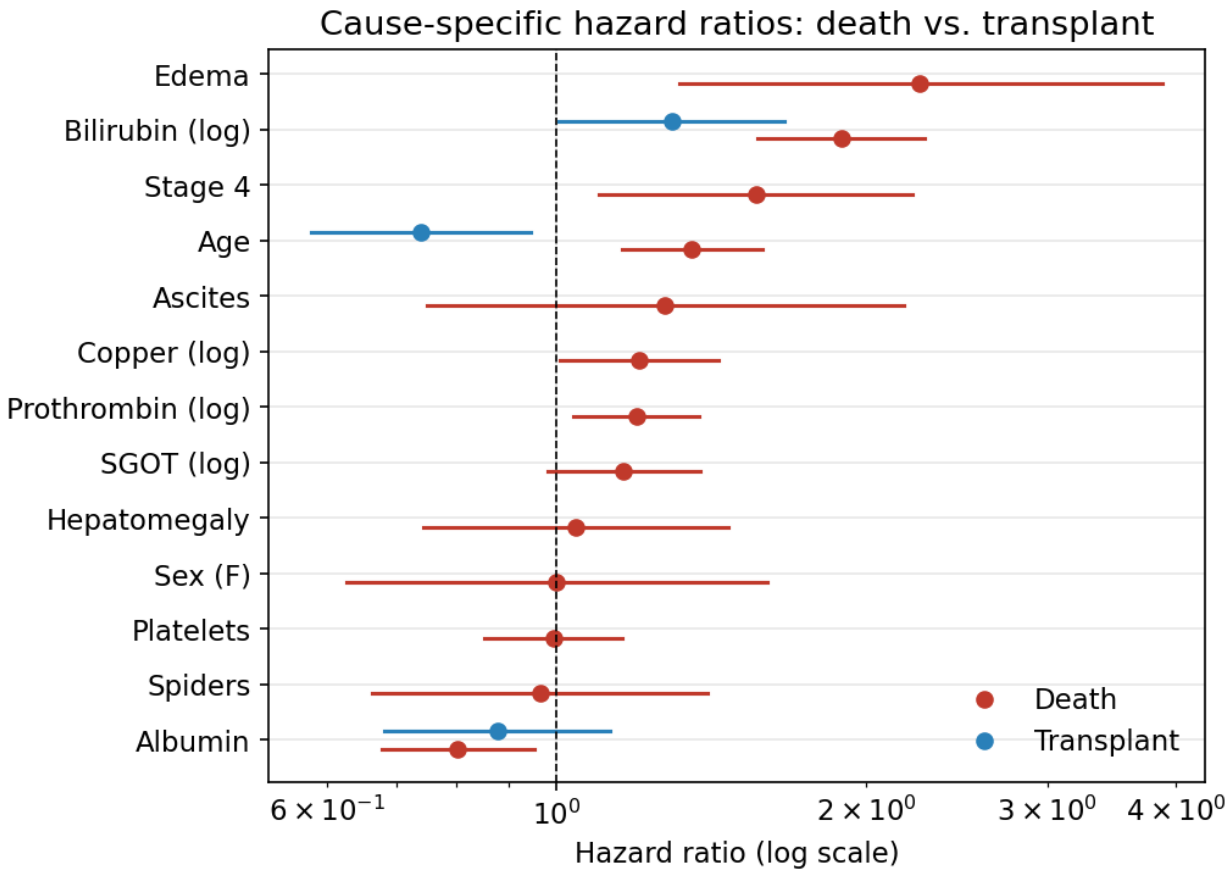


Figure S3: Cause-specific hazard ratios for death (red) and transplantation (blue). Age sits above 1 for death but below 1 for transplantation, the opposite-direction effect. (Companion to the cause-specific Cox table.)

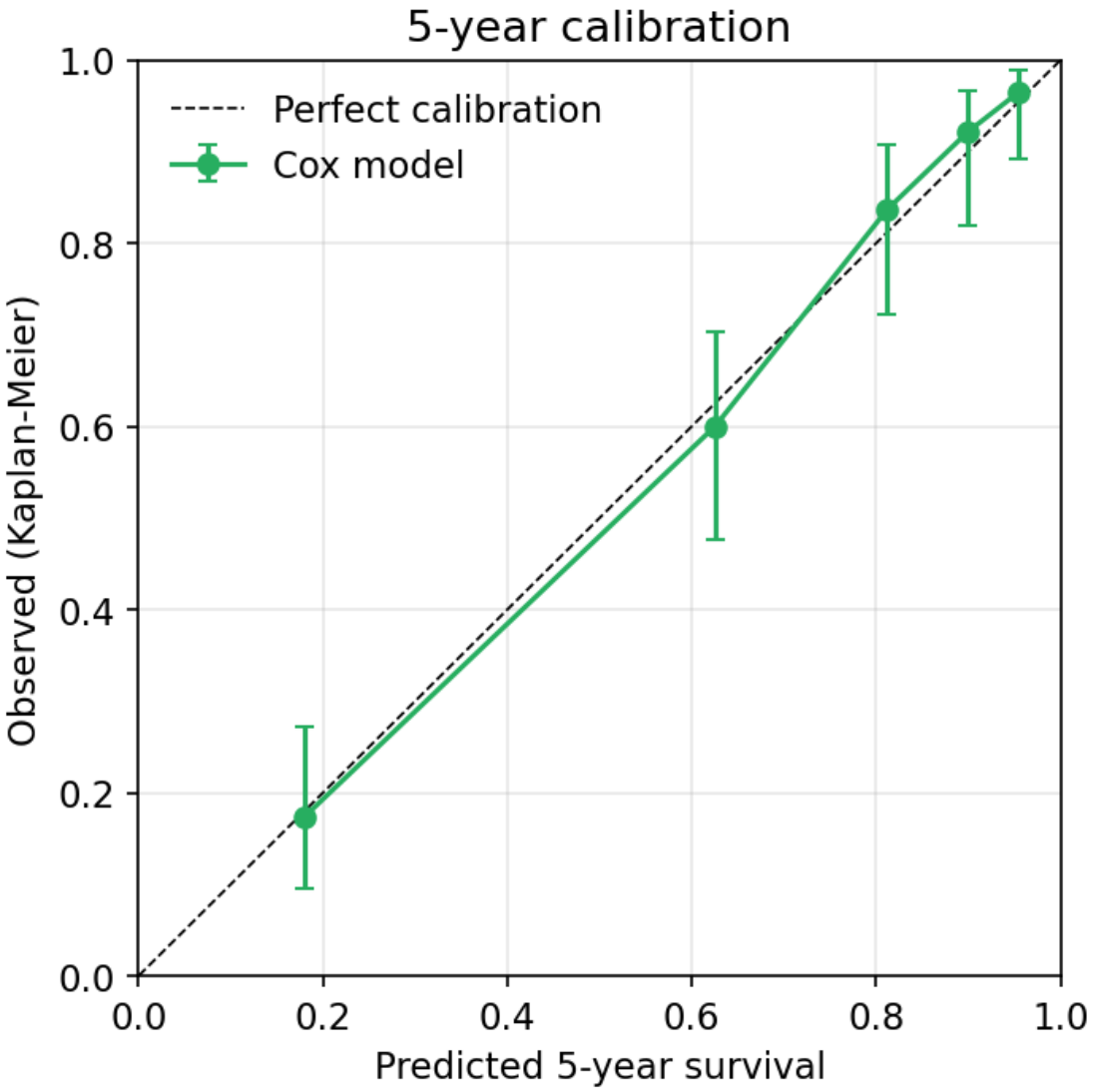


Figure S4: Five-year calibration of the Cox model: predicted survival against Kaplan–Meier observed survival with transplant censored, by risk quintile.

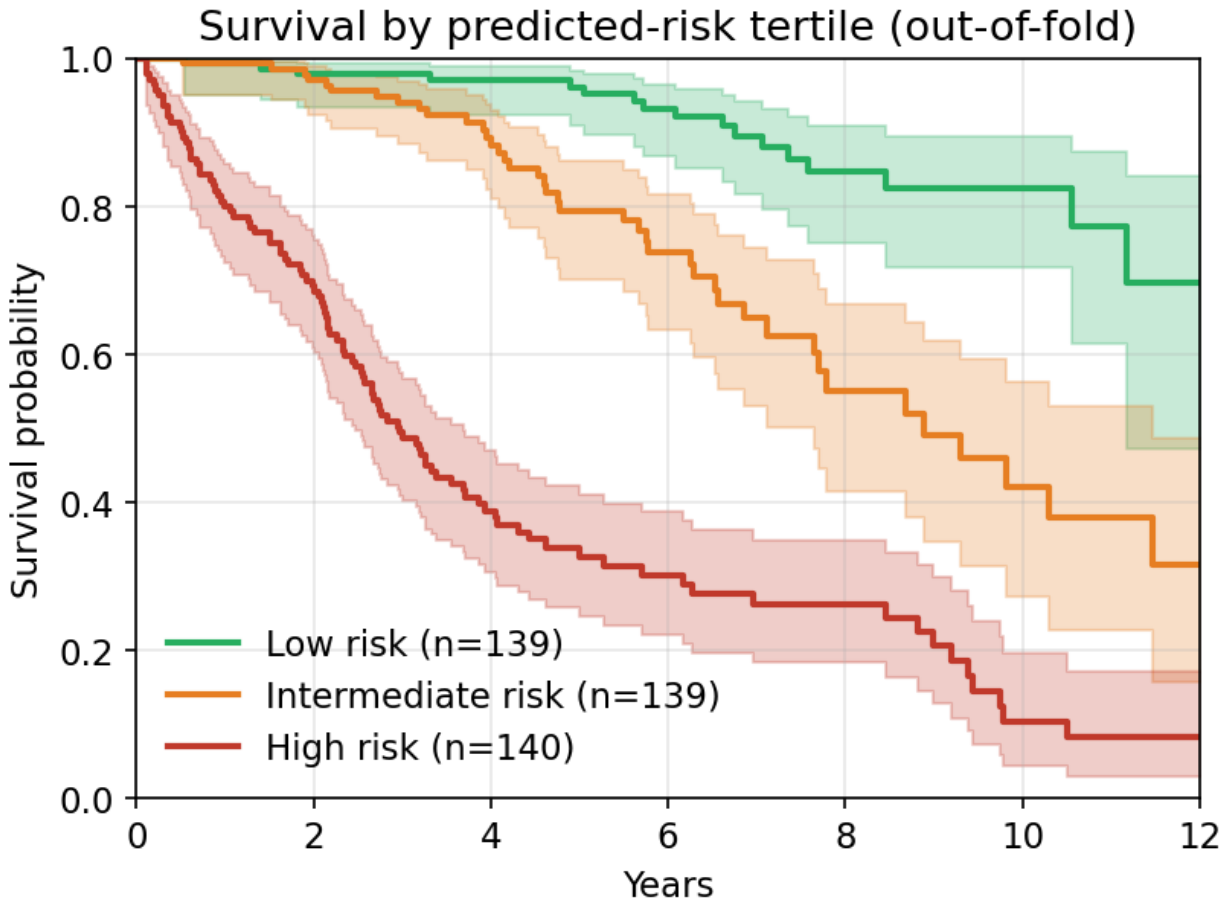


Figure S5: Kaplan–Meier survival by out-of-fold predicted-risk tertile.

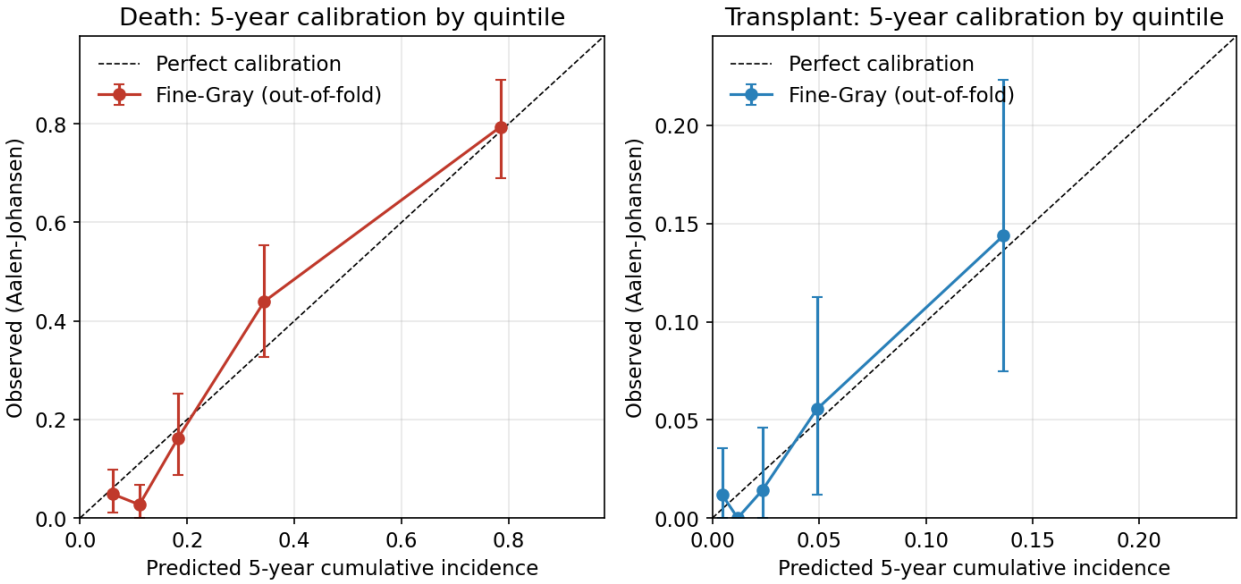


Figure S6: Fine–Gray five-year calibration: mean out-of-fold predicted cumulative incidence against Aalen–Johansen observed incidence (95% bootstrap interval) by predicted-risk quintile, for death (left) and transplant (right). (Companion to the Fine–Gray calibration paragraph in the main text.)